Data Centre Uptime: Operating Practices

What operators of Tier III facilities do every day, week and quarter

Uptime in a data centre is mostly the result of discipline: knowing what changed, rehearsing failure and respecting maintenance windows. These practices come from the Data Centre Uptime Clinic held by the Hyderabad Chapter, and apply to enterprise rooms as well as colocation facilities.

1. Daily and weekly

- Walk-through with checklist: leaks, alarms, temperatures, fuel levels, battery status
- Review the last 24 hours of alarms and the changes scheduled for the next 24
- Verify generator auto-start readiness and fuel above 80% of tank capacity
- Weekly test of the building management alarms to the on-call phone

2. Change management

1. Every change has a written method of procedure (MOP) with step-by-step actions, rollback and abort criteria.
2. Changes to critical power or cooling need two engineers and a second reviewer.
3. Freeze windows are published; no non-urgent change during quarter-end or major customer events.
4. After the change, verify and log the as-found and as-left readings.

3. Testing

Test	Frequency	Pass criterion
Generator on load	Monthly, 30 minutes	Starts in under 10 seconds; stable voltage
UPS battery discharge	Quarterly	At least 90% of rated autonomy
Chiller changeover	Quarterly	No rise above 1.5 deg C in cold aisle
Full building transfer	Annually	ATS transfer without IT impact
Fire suppression and EPO drill	Annually	Procedure followed; no accidental release

4. Metrics that matter

- Mean time between failures by system
- PUE trend with load, reported monthly
- Change failure rate (target under 2%)
- Incident post-mortems completed within five days